MANCHIDI MAGASENG JOHN

MSc Data Engineering Student | Python · SQL · PySpark · Machine Learning

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PROFESSIONAL SUMMARY

Computer Science graduate (BSc Honours) and MSc Data Engineering student at Constructor University in Bremen, with project experience in Python, SQL, PySpark and data pipeline development. Experience in big data processing, machine learning (scikit-learn, TensorFlow, OpenCV) and building REST APIs and analytics applications (FastAPI, Streamlit, Docker). Mathematical and statistical foundation from a BSc in Mathematical Sciences. Seeking junior and working-student roles in data engineering, data analytics and software development in Germany.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, Java, C++, Bash, JavaScript (basic)

Data Engineering & Analytics: PySpark, ETL / data pipelines, big data processing, Pandas, data cleaning and validation, feature engineering, statistical modelling, Plotly, Excel (pivot tables), Power BI (basic)

Machine Learning & AI: scikit-learn, TensorFlow, OpenCV, classification, model evaluation, data-leakage auditing, agentic AI workflows

Databases: PostgreSQL, MySQL, relational schema design, SQL query optimisation

Cloud & DevOps: AWS, Docker, Git, GitHub, Linux

Software Development: REST APIs, FastAPI, Flask, Streamlit, HTML / CSS

Cybersecurity & Networking: Snort (IDS/IPS), pfSense, Wireshark, Nmap, Nessus, Metasploit, Active Directory

TECHNICAL PROJECTS

E-Commerce Customer Churn & Big Data Analytics 2026

Individual academic project | Python, PySpark, scikit-learn, Pandas, Jupyter

- Built a reproducible PySpark data-processing pipeline to clean, transform and engineer features from e-commerce customer data.
- Segmented customers with RFM analysis and trained scikit-learn classification models to predict churn.
- Audited the modelling workflow for data leakage and documented the findings, so results reflect a leakage-aware evaluation.

InspectAI: Agentic Visual Quality Inspector 2026

Personal project (competition entry) | Python, OpenCV, FastAPI, Streamlit, Docker, AWS

- Developed a computer-vision application that localises visual anomalies on manufactured surfaces using OpenCV.
- Implemented an agentic decision loop (re-inspect, pass, flag or escalate to a human) with a fully auditable decision trace.
- Exposed the workflow through a FastAPI REST backend and a Streamlit interface, using Docker and AWS.

FinTrack SA: Personal Finance Tracker 2026

Personal project | Python, Streamlit, Plotly

- Built a Streamlit application (amounts in ZAR) for transaction and budget management, account balances and savings goals.
- Created interactive Plotly dashboards to visualise spending and budget performance.

Adaptive Ransomware Detection (BSc Honours Thesis) 2025

University of Limpopo | Python, scikit-learn, TensorFlow, Pandas

- Evaluated supervised and unsupervised machine-learning models for ransomware detection on public datasets (Malimg, Microsoft BIG 2015).
- Best Random Forest model reached 99.34% accuracy on the evaluated dataset; analysed feature importance and adaptability to unseen variants. This is a single-dataset result, not a claim of real-world detection performance.

Secure Backend Database System 2025

University of Limpopo, Advanced Database | Python, MySQL, PostgreSQL

- Designed relational schemas in MySQL and PostgreSQL for secure storage and reliable reporting.
- Optimised SQL queries and carried out data validation, integrity testing and debugging.

FURTHER TECHNICAL PROJECTS

Statistical Modelling & Data Analysis 2023
University of Limpopo, group project | R, Python

- Applied regression and statistical modelling to datasets to identify patterns and produce predictive insights.
- Presented findings and recommendations to project stakeholders through structured reports and visualisations.

Intrusion Detection & Prevention with Snort Oct 2025
University of Limpopo, Computer Security | Snort, Kali Linux, Ubuntu, Metasploitable 2, VirtualBox

- Built an isolated VirtualBox lab (Kali attacker, Ubuntu Snort sensor, Metasploitable 2 target) to test network attack detection.
- Wrote custom Snort rules for port scans and well-known exploit signatures; documented the work in a 66-page report.

Cybersecurity HomeLab (SME-style Network) Jan – Jun 2025
Personal / University of Limpopo, Computer Networking | pfSense, Windows Server, Active Directory, Nessus

- Built a segmented virtual network with a pfSense firewall and a Windows Server domain controller (AD DS, DNS, DHCP).
- Ran Nessus vulnerability scans from a Kali workstation; documented the design and implementation in an 18-page report.

EDUCATION

MSc Data Engineering, *Constructor University, Bremen, Germany* Sep 2026 – present
Focus: Large-scale data systems, data pipelines and the engineering foundations of data-driven applications

MSc Computer Science (dissertation), *University of Limpopo, South Africa* Jan 2026 – present
Focus: Dissertation building on honours research in machine learning and security

BSc Honours, Computer Science, *University of Limpopo, South Africa* 2025
Thesis: Evaluating machine learning techniques for adaptive ransomware detection
Coursework: Advanced Database, Advanced Computer Networks, Cybersecurity, Advanced Operating Systems

BSc Mathematical Sciences (Computer Science and Statistics), *University of Limpopo, South Africa* 2024
Coursework: Algorithms, Data Structures, Database Systems, Regression Analysis, Time Series Analysis, Artificial Intelligence, Sampling Theory, Experimental Design

National Senior Certificate (Matric), *Letebele Secondary School, South Africa* 2019

CERTIFICATIONS & ADDITIONAL PROGRAMMES

- **Cisco Cybersecurity Essentials**, Cisco Networking Academy
- **Cisco Data Science Essentials**, Cisco Networking Academy
- **Centre for High Performance Computing (CHPC) Programme**, NITheCS

LANGUAGES

English: Fluent (language of study) **German:** Beginner, currently learning